Analysis I

Uebungsblatt 2

Problem:	Comment:	Points:
1		28.5 / 32

1

(a)

Let $c \geq 0$. We claim: $|a| \leq c \iff -c \leq a \leq c$.

Proof. By the definition of absolute value, for every a we have

$$-|a| \leq a \leq |a|.$$

(Indeed, if $a \geq 0$ then $|a| = a$; if $a < 0$ then $|a| = -a$.)

($\Rightarrow$) If $|a| \leq c$, then

$$-c \leq -|a| \leq a \leq |a| \leq c,$$

so $-c \leq a \leq c$.

($\Leftarrow$) If $-c \leq a \leq c$, then: - If $a \geq 0$, $|a| = a \leq c$. - If $a \leq 0$, then $0 \leq -a \leq c$, hence $|a| = -a \leq c$.

Thus $|a| \leq c \iff -c \leq a \leq c$. 2

(b) Triangle inequality

From (a) with $c = |a|$ and $c = |b|$ we have

$$-|a| \leq a \leq |a|, \qquad -|b| \leq b \leq |b|.$$

Adding these inequalities gives

$$-(|a| + |b|) \leq a + b \leq |a| + |b|.$$

Since $|a| + |b| \geq 0$, applying (a) again (with $x = a + b$ and $c = |a| + |b|$) yields

$$|a + b| \leq |a| + |b|.$$

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(c) Reverse triangle inequality

Apply (b) to $a = (a - b) + b$:

$$|a| \leq |a - b| + |b| \quad \Rightarrow \quad |a| - |b| \leq |a - b|.$$

Swapping a and b gives

$$|b| \leq |a - b| + |a| \quad \Rightarrow \quad |b| - |a| \leq |a - b|.$$

Combining both we get

$$||a| - |b|| \leq |a - b|.$$

2

(d) Triangle inequality with a midpoint

Note that $a - c = (a - b) + (b - c)$. By (b),

$$|a - c| = |(a - b) + (b - c)| \leq |a - b| + |b - c|.$$

2

2

Determine the infimum and supremum of the set

$$M := \left\{ \frac{m-n}{m+n} : m, n \in \mathbb{N},\ n \neq 0 \right\}.$$

Solution. Fix $m, n \in \mathbb{N}$ with $n \geq 1$ (so $m + n > 0$). Set

$$x := \frac{m-n}{m+n}.$$

Then the following identities give immediate bounds:

$$1 - x = 1 - \frac{m-n}{m+n} = \frac{2n}{m+n} > 0 \quad \Rightarrow \quad x < 1,$$

and

$$x + 1 = \frac{m-n}{m+n} + 1 = \frac{2m}{m+n} \geq 0 \quad \Rightarrow \quad x \geq -1,$$

with equality $x = -1$ iff $m = 0$. Hence for all admissible m, n we have

$$-1 \leq \frac{m-n}{m+n} < 1.$$

Supremum. For every $x \in M$ we just saw $x < 1$, so 1 is an upper bound. To see it is the least upper bound, fix $\varepsilon > 0$ and let $n = 1$, $m \in \mathbb{N}$ large. Then

$$\frac{m-1}{m+1} = 1 - \frac{2}{m+1} > 1 - \varepsilon$$

whenever $m + 1 > 2/\varepsilon$. Thus elements of M come arbitrarily close to 1 from below, and therefore

$$\sup M = 1.$$

Moreover, no element equals 1 since $n \geq 1$ forces $1 - x = 2n/(m+n) > 0$.

Infimum. From $x \geq -1$ we have -1 is a lower bound. If $0 \in \mathbb{N}$ is allowed (as is common in this course and consistent with the condition $n \neq 0$), choose $m = 0$ and any $n \geq 1$ to get $x = -1 \in M$. Hence

$$\inf M = -1, \quad \text{and } \min M = -1 \text{ is attained (at } m = 0).$$

If, alternatively, one uses $\mathbb{N} = \{1, 2, \ldots\}$, then $m \geq 1$ and equality $x = -1$ is not attained. However, taking $m = 1$ and $n \to \infty$ gives

$$\frac{1-n}{1+n} = -1 + \frac{2}{1+n} \searrow -1,$$

so still $\inf M = -1$ but there is no minimum. $\square$

3

3

Let $A \subset \mathbb{R}$ with $\alpha := \inf A > 0$, and set $A^{-1} := \{1/a : a \in A\}$. We prove $\sup(A^{-1}) = 1/\alpha$ by the "no smaller upper bound" method.

Step 1: $1/\alpha$ is an upper bound of A^{-1}. For every $a \in A$ we have $\alpha \leq a$ and $\alpha > 0$. Since inversion is order-reversing on $(0, \infty)$,

$$\frac{1}{a} \leq \frac{1}{\alpha}.$$

Hence $1/\alpha$ is an upper bound of A^{-1}.

Step 2: No smaller number is an upper bound. Let $u < 1/\alpha$.

- If $u \leq 0$, pick any $a \in A$; then $1/a > 0 \geq u$, so u is not an upper bound.
- If $u > 0$, then $u < 1/\alpha$ implies $1/u > \alpha$. Define $\varepsilon := 1/u - \alpha > 0$. By the definition of inf, there exists $a \in A$ with

  $$\alpha \leq a < \alpha + \varepsilon = \frac{1}{u}.$$

  As $a > 0$, inversion reverses inequalities:

  $$\frac{1}{a} > \frac{1}{\alpha + \varepsilon} = u.$$

  Thus there is an element of A^{-1} exceeding u, so u is not an upper bound.

Since $1/\alpha$ is an upper bound and every smaller number fails to be one, $1/\alpha = \sup(A^{-1})$. C

4

(a)

Yes. $\mathbb{N}$ and $\mathbb{Q}$ are equinumerous (both are countably infinite).

Construct an explicit enumeration of $\mathbb{Q}$. List 0 first. For each integer $s = 2, 3, 4, \ldots$, traverse the diagonal

$$\{(p, q) \in \mathbb{N}^2_{\geq 1} : p + q = s\}$$

in increasing p, and append to the list the two rationals

$$\frac{p}{q} \quad \text{and} \quad -\frac{p}{q}$$

but only when $\gcd(p, q) = 1$ (i.e., the fraction is in lowest terms).

This produces an infinite sequence

$$0,\ 1, -1,\ \tfrac{1}{2}, -\tfrac{1}{2},\ 2, -2,\ \tfrac{1}{3}, -\tfrac{1}{3},\ \tfrac{2}{3}, -\tfrac{2}{3},\ \ldots$$

in which: - Surjectivity: Every $r \in \mathbb{Q}$ has a unique reduced form $r = \pm\frac{m}{n}$ with $m \in \mathbb{N}_{\geq 0}$, $n \in \mathbb{N}_{\geq 1}$, $\gcd(m,n) = 1$. Then (m,n) lies on the diagonal $m + n = s$, so r appears in the list. - Injectivity: We only list reduced fractions, and each sign is listed once, so no rational appears twice.

Let $f : \mathbb{N} \to \mathbb{Q}$ map k to the k-th term of this sequence. Then f is bijective, proving $\mathbb{N} \sim \mathbb{Q}$.

2

(b)

Let $[a,b], [c,d] \subset \mathbb{R}$.

Case 1: $a < b$ and $c < d$ (non-degenerate intervals). Define the affine map

$$f : [a,b] \to [c,d], \qquad f(x) := c + \frac{d-c}{b-a}(x-a).$$

Then $f(a) = c$ and $f(b) = d$, and f is strictly increasing (hence injective). For any $x \in [a,b]$ we can write $x = a + t(b-a)$ with $t \in [0,1]$, so

$$f(x) = c + t(d-c) = (1-t)c + td \in [c,d],$$

showing f is surjective as well. The inverse is the affine map

$$f^{-1}(y) = a + \frac{b-a}{d-c}(y-c), \qquad y \in [c,d].$$

Therefore f is a bijection; the intervals are equinumerous.

Case 2: $a = b$ and $c = d$ (both singletons). The map $f(a) = c$ is a bijection.

Remark (edge case): If exactly one interval is a singleton (e.g., $a = b$ but $c < d$), then the intervals are not equinumerous, since one has cardinality 1 and the other has the cardinality of the continuum.

3

(c)

Yes. $(0,1)$ is equinumerous with $\mathbb{R}$.

Define

$$f : (0,1) \longrightarrow \mathbb{R}, \qquad f(x) := \tan\big(\pi(x - \tfrac{1}{2})\big).$$

No derivatives or limits yet

Then f is strictly increasing because $\cancel{f'(x) = \pi \sec^2\big(\pi(x - \tfrac{1}{2})\big)} > 0$ on $(0,1)$, hence injective. Moreover,

$$\cancel{\lim_{x \to 0^+} f(x) = \tan\big(-\tfrac{\pi}{2}\big) = -\infty, \qquad \lim_{x \to 1^-} f(x) = \tan\big(\tfrac{\pi}{2}\big) = +\infty,}$$

so f is surjective onto $\mathbb{R}$. The inverse map is explicit:

$$f^{-1}(y) = \tfrac{1}{2} + \frac{1}{\pi}\arctan(y), \qquad y \in \mathbb{R}.$$

Thus f is a bijection and $(0,1) \sim \mathbb{R}$ (both have cardinality of the continuum).

1

5

(a)

Let $a \in \mathbb{R}$ with $a > 0$ and $p \in \mathbb{N}$. Set

$$B := \{x \in \mathbb{R} : \ x \geq 0, \ x^p \leq a\}.$$

We show that B is bounded above.

If $a \geq 1$, then for every $x \geq a$ we have $x^p \geq x \geq a$ (since $p \geq 1$ and $x \mapsto x^p$ is increasing on $[0, \infty)$), hence $x \notin B$. Therefore a is an upper bound of B.

If $0 < a < 1$, then for every $x \geq 1$ we have $x^p \geq 1 > a$, so $x \notin B$. Hence 1 is an upper bound of B.

Thus in all cases B is bounded above, with the explicit bound

$$\sup B \leq M, \qquad M := \max\{1, a\}.$$

3

5(b)

Let $B = \{x \in \mathbb{R} : \ x \geq 0, \ x^p \leq a\}$ with $a > 0$, $p \in \mathbb{N}$, and let $c := \sup B$. We prove $c^p = a$.

Claim (hint). For every $p \in \mathbb{N}$ and every ε with $0 < \varepsilon < c$ there exists a constant $R = R(p, c) \geq 0$ (depending only on p and c, not on ε) such that

$$(c+\varepsilon)^p \leq c^p + \varepsilon R, \qquad (c-\varepsilon)^p \geq c^p - \varepsilon R. \tag{$\star$}$$

Proof of the claim (by induction on p). For $p = 1$, take $R = 1$, and equalities hold. Suppose $(\star)$ holds for p with some $R_p = R(p, c)$. Then for $p + 1$,

$$(c+\varepsilon)^{p+1} = (c+\varepsilon)^p(c+\varepsilon) \leq (c^p + \varepsilon R_p)(c+\varepsilon) = c^{p+1} + \varepsilon(R_p c + c^p) + \varepsilon^2 R_p.$$

Since $0 < \varepsilon < c$, we have $\varepsilon^2 R_p \leq \varepsilon\, c\, R_p$. Hence

$$(c+\varepsilon)^{p+1} \leq c^{p+1} + \varepsilon\big(R_p c + c^p + cR_p\big) = c^{p+1} + \varepsilon R_{p+1},$$

with $R_{p+1} := c^p + 2cR_p$, which depends only on $p+1$ and c. Similarly,

$$\begin{aligned}
(c-\varepsilon)^{p+1} &= (c-\varepsilon)^p(c-\varepsilon) && (1)\\
&\geq (c^p - \varepsilon R_p)(c-\varepsilon) && (2)\\
&= c^{p+1} - \varepsilon(R_p c + c^p) + \varepsilon^2 R_p && (3)\\
&\geq c^{p+1} - \varepsilon(c^p + cR_p) && (4)\\
&= c^{p+1} - \varepsilon R'_{p+1}, && (5)
\end{aligned}$$

where $R'_{p+1} := c^p + cR_p \leq c^p + 2cR_p =: R_{p+1}$. Enlarging the constant if necessary, the same R_{p+1} works for both inequalities. This completes the induction and proves the claim.

Step 1: $c^p \geq a$. By definition of $c = \sup B$, for every $\varepsilon \in (0, c)$ we have $c - \varepsilon \in B$ (otherwise $c - \varepsilon$ would be an upper bound of B smaller than c). Hence

no lines

$$(c - \varepsilon)^p \leq a.$$

Using $(\star)$,

$$c^p - \varepsilon R \leq (c - \varepsilon)^p \leq a \quad \Rightarrow \quad c^p \leq a + \varepsilon R.$$

Letting $\varepsilon \downarrow 0$ yields $c^p \leq a$. Wait—this gives the reverse inequality. So we instead argue as follows: since c is an upper bound of B, every $x \in B$ satisfies $x \leq c$ and hence $x^p \leq c^p$, from which $\sup\{x^p : x \in B\} \leq c^p$. But $\{x^p : x \in B\} = [0, a]$ by definition of B, so $a \leq c^p$. Thus $c^p \geq a$.

Step 2: $c^p \leq a$. For every $\varepsilon > 0$, the number $c + \varepsilon$ is not an upper bound of B; hence there exists $x_\varepsilon \in B$ with $x_\varepsilon > c - \frac{\varepsilon}{2}$. Then

$$c < x_\varepsilon + \tfrac{\varepsilon}{2} \leq c + \varepsilon,$$

so by monotonicity on $[0, \infty)$,

$$c^p < (x_\varepsilon + \tfrac{\varepsilon}{2})^p \leq (c + \varepsilon)^p.$$

Using $(\star)$,

$$c^p < (c + \varepsilon)^p \leq c^p + \varepsilon R,$$

hence $0 < (c + \varepsilon)^p - c^p \leq \varepsilon R$. Letting $\varepsilon \downarrow 0$ gives

2.5

$$\lim_{\varepsilon \downarrow 0} (c + \varepsilon)^p = c^p.$$

But since $x_\varepsilon \in B$, we have $x_\varepsilon^p \leq a$, and from $x_\varepsilon > c - \frac{\varepsilon}{2}$ and $(\star)$ (applied with $\varepsilon/2$) we get

$$c^p - \tfrac{\varepsilon}{2} R \leq (c - \tfrac{\varepsilon}{2})^p \leq x_\varepsilon^p \leq a.$$

Letting $\varepsilon \downarrow 0$ yields $c^p \leq a$.

Combining Steps 1 and 2 we conclude $c^p = a$.

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